

Course Six

Google Advanced Data Analytics Capstone



Instructions

Use this PACE strategy document to record your decisions and reflections as a data professional as you work through the capstone project. As a reminder, this document is a resource guide that you can reference in the future and a space to help guide your responses and reflections posed at various points throughout the project.

Portfolio Project Recap

Many of the goals you accomplished in your individual course portfolio projects are incorporated into the Advanced Data Analytics capstone project including:

- Create a project proposal
- Demonstrate understanding of the form and function of Python
- Show how data professionals leverage Python to load, explore, extract, and organize information through custom functions
- Demonstrate understanding of how to organize and analyze a dataset to find the “story”
- Create a Jupyter notebook for exploratory data analysis (EDA)
- Create visualization(s) using Tableau
- Use Python to compute descriptive statistics and conduct a hypothesis test
- Build a multiple linear regression model with ANOVA testing
- Evaluate the model
- Demonstrate the ability to use a notebook environment to create a series of machine learning models on a dataset to solve a problem
- Articulate findings in an executive summary for external stakeholders



Project proposal

Salifort Motors Employee Attrition project proposal

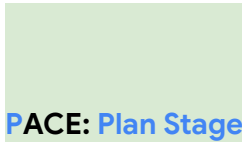
Overview

Salifort's senior leadership team has requested a model that uses data from an employee survey to predict employee departure from the company

Milestones	Tasks	Status	PACE stages
1	Write a Project Proposal and Draft Research Question	Completed ▾	Plane
2	Compile Summary Information + Exploratory Data Analysis	Completed ▾	Plan, Analyze
3	Data Cleaning, Transformation and Processing	Completed ▾	Analyze
4	Descriptive Statistics and Data Visualization	Completed ▾	Analyze
5	Build a Logistic Regression Model and Evaluate it	Completed ▾	Analyze, Construct
6	Build Machine Learning Models and Evaluate Them	Completed ▾	Analyze, Construct
7	Share Final Insights with Stakeholders	Completed ▾	Execute



Data Project Questions & Considerations



PACE: Plan Stage

Foundations of Data Science

- Who is your audience for this project?
 - [Salifort Leadership Team, HR Team, Data Analyst Team](#)
- What are you trying to solve or accomplish? And, what do you anticipate the impact of this work will be on the larger business need?
 - [The goal is to analyze the data collected by the HR department and to build a model that predicts whether or not an employee will leave the company. The current high turnover rate is costly in the financial sense as Salifort makes a big investment in recruiting, training, and upskilling its employees. By predicting employee turnover and understanding the root causes of departures, we can pinpoint contributing factors. This insight allows us to develop solutions that boost employee retention and increase job satisfaction, ultimately saving the company time and money.](#)
- What questions need to be asked or answered?
 - [What kind of employees are leaving? Are there any trends? What are the satisfaction and performance levels of those who leave vs those who don't? What are potential contributing factors behind job attrition? What is the average tenure of those who leave vs those who don't?](#)
- What resources are required to complete this project? [Coursera Course Materials and Labs: Videos, reading assignments, and glossaries. Python and Python Documentation. Gemini](#)
- What are the deliverables that will need to be created over the course of this project? [See Project Plan above](#)
- How can you best prepare to understand and organize the provided information? [Review Labs and Notebooks](#)
- What follow-along and self-review codebooks will help you perform this work? [The ones for Waze Project + Those included in the program](#)
- What are a couple additional activities a resourceful learner would perform before starting to code? [Learn more about how to predict attrition](#)



Go Beyond the Numbers: Translate Data into Insights

- What are the data columns and variables and which ones are most relevant to your deliverable?
 - left (Whether or not the employee left the company, Y variable)
 - **satisfaction_level** (The employee's self-reported satisfaction level [0-1])
 - **last_evaluation** (Score of employee's last performance review [0-1])
 - number_project (Number of projects employee contributes to)
 - average_monthly_hours (Average number of hours employee worked per month)
 - **time_spend_company** (How long the employee has been with the company (years))
 - work_accident (Whether or not the employee experienced an accident while at work)
 - **promotion_last_5years** (Whether or not the employee was promoted in the last 5 years)
 - department (The employee's department)
 - **salary** (The employee's salary (low, medium, or high))
- What units are your variables in?
 - left: Binary Variable: (0-1), no units
 - satisfaction_level: Range of 0 to 1, no units
 - last_evaluation: Range of 0 to 1, no units
 - number_project: Integer Number, no units
 - average_monthly_hours: Hours
 - time_spend_company: Years
 - work_accident: Binary Variable: (0-1), no units
 - promotion_last_5years: Binary Variable: (0-1), no units
 - department: : Categorical String Variable,
 - salary: Categorical String Variable, no units
- What are your initial presumptions about the data that can inform your EDA, knowing you will need to confirm or deny with your future findings?
 - Salary, promotion_last_5years, satisfaction_level, time_spend_company, last_evaluation are expected to be the strongest predictors
- Is there any missing or incomplete data?
 - No
- Are all pieces of this dataset in the same format?



- No
- Which EDA practices will be required to begin this project?
 - Ask
 - Prepare
 - Process
 - Analyze
 - Share

The Power of Statistics

- What is the main purpose of this project?
 - Save Time and Money by determining reasons behind employee departure in order to increase employee retention and job satisfaction
- What is your research question for this project?
 - What are the contributing factors behind an employee leaving the company?
- What is the importance of random sampling? In this case, what is an example of sampling bias that might occur if you didn't use random sampling?
 - Random Sampling ensures that our sample is representative of the population as a whole i.e, it accurately reflects the characteristics of a population . This helps reduce sampling bias, increases the validity of our results and enables us to make valid inferences or accurate predictions about a population .

Regression Analysis: Simplify Complex Data Relationships

- Who are your stakeholders for this project?
 - Salifort Leadership Team, HR Team, Data Analyst Team
- What are you trying to solve or accomplish?
 - What are the contributing factors behind an employee leaving the company? Which of these are statistically significant?
- What are your initial observations when you explore the data?



- Many important variables are present, however some are missing e.g gender (are women or men more likely to leave), age (are younger or older employees likely to leave), Did the employee receive a bonus? Are parents more likely to leave?. Data from outside sources e.g. average market salary for employees role and department and how the current salary compares to it. Data from exit interviews: did they have a counter offer? ,Employee geographic location (to compare market conditions such as average salary)? Benefits offered?
- No missing data
- Many duplicates
- All employees worked a minimum of 2 projects for the company and a maximum of 7. The mean is around 4
- All employees worked a minimum of 96 hours each month on average and a maximum of 310. The mean is around 201 which is quite higher than the industry average of 171 for USA ($42.8 * 4$).
Source: [Automotive Industry: Employment, Earnings, and Hours](#)
- All employees stayed a minimum of two years for the company and a maximum of 10 years. The mean was 3.5 years
- Most respondents didn't face an accident
- Most respondents didn't leave
- Most employees didn't get promoted in the last 5 years.
- What resources do you find yourself using as you complete this stage? (Make sure to include the links.)
Coursera Course Materials and Labs: Videos, reading assignments, and glossaries. Python and Python Documentation. Gemini
- Do you have any ethical considerations at this stage?
 - Is there any PII, or personally identifiable information? Has PII been eliminated and are we looking at aggregate information?
 - Were Data Privacy measures implemented and was it protected? Was a data subject's information and activity preserved
 - Was the consent of individuals asked for before they were surveyed? Did employees consent to their data being used for models, such as this one, that help drive policy? Did employees choose how their data is used?



The Nuts and Bolts of Machine Learning

- What am I trying to solve?
 - What are the contributing factors behind an employee leaving the company? Which of these are the most important or best parameters in predicting which employees are likely to leave?
- What resources do you find yourself using as you complete this stage?
 - Coursera Course Materials and Labs: Videos, reading assignments, and glossaries. Python and Python Documentation. Gemini. VScode Intellisense
- Is my data reliable?
 - Yes as it seems to have key variables without any noticeable bias
- Do you have any additional ethical considerations in this stage?
 - Are there any potential biases about the type of people who leave? Anything related to gender, nationality, religion or race? Is the data biased? How do we avoid bias being injected and reproduced in our models
- What data do I need/would I like to see in a perfect world to answer this question?
 - Many important variables are present, however some are missing e.g gender (are women or men more likely to leave), age (are younger or older employees likely to leave), Did the employee receive a bonus? Are parents more likely to leave?. Data from outside sources e.g. average market salary for employees role and department and how the current salary compares to it, How do the benefits offered by the company compare to the industry?, Employee geographic location (to compare market conditions such as average salary). Data from exit interviews: did they have a counter offer? Any temporal or seasonal features (Year of data collection/month) etc?
- What data do I have/can I get?
 - satisfaction_level, last_evaluation , number_project , average_monthly_hours
time_spend_company , work_accident, promotion_last_5years , department , salary
 - new data:
 - Proportion of Employees in the critical retention period of 3-5
- What metric should I use to evaluate success of my business objective? Why?
 - F1 Score. In this analysis it is important to have high quality true positives. F1 score balances Precision and recall and so is a good candidate for this project



Data Project Questions & Considerations



PACE: Analyze Stage

Foundations of Data Science

- Will the available information be sufficient to achieve the goal based on your intuition and the analysis of the variables?
 - Not enough for a full analysis but good enough to start

Go Beyond the Numbers: Translate Data into Insights

- What steps need to be taken to perform EDA in the most effective way to achieve the project goal?
 - Discovering
 - Structuring
 - Cleaning
 - Joining
 - Validating
 - Presenting
- Do you need to add more data using the EDA practice of joining? What type of structuring needs to be done to this dataset, such as filtering, sorting, etc.?
 - No
 - Filtering
 - Encoding
- What initial assumptions do you have about the types of visualizations that might best be suited for the intended audience?
 - Histograms
 - BoxPlots
 - Lineplots
 - Scatterplots



- Heatmap

The Power of Statistics

- Why are descriptive statistics useful?
 - In general, descriptive statistics are useful because they let you quickly explore and understand large amounts of data.
- What is the difference between the null hypothesis and the alternative hypothesis?
 - The null hypothesis is a statement that is assumed to be true unless there is convincing evidence to the contrary. The null hypothesis typically assumes that there is no effect in the population, and that your observed data occurs by chance.
 - The alternative hypothesis is a statement that contradicts the null hypothesis, and is accepted as true only if there is convincing evidence for it. The alternative hypothesis typically assumes that there is an effect in the population, and that your observed data does not occur by chance.

Regression Analysis: Simplify Complex Data Relationships

- What are some purposes of EDA before constructing a multiple linear regression model?
 - Check distributions
 - Check for descriptive statistics to understand the data
 - Check model assumptions
 - Check for relationships
 - Check for any errors like duplicates and missing values
- Do you have any ethical considerations in this stage?
 - The current dataset lacks several key variables that could provide a more comprehensive view of employee turnover (,Demographics, Upward feedback, Total Rewards,Market Benchmarks, Geographic Context, Exit Insights, Temporal and Seasonal Features)
 - The absence of critical variables introduces a risk of omitted variable bias, which can lead to skewed results or the misrepresentation of specific employee cohorts. Specifically, the lack of demographic data makes it impossible to verify the representativeness of the sample regarding gender parity or age distribution, potentially limiting the model's generalizability.
 -

- The absence of temporal and seasonal features limits the granularity of the data, hindering our ability to perform longitudinal analyses or isolate specific seasonal drivers of attrition.

The Nuts and Bolts of Machine Learning

- What am I trying to solve? Does it still work? Does the plan need revising?
 - What are the contributing factors behind an employee leaving the company? Which of these are the most important or best parameters in predicting which employees are likely to leave?
 - It still works
- Does the data break the assumptions of the model? Is that ok, or unacceptable?
 - Logistic Regression Linearity Assumptions violated
 - We moved to a different model
- Why did you select the X variables you did?
 - I selected the X variables most likely expected to have an impact on the data
- What are some purposes of EDA before constructing a model?
 - Check distributions
 - Check for descriptive statistics to understand the data
 - Check model assumptions
 - Check for relationships
 - Check for any errors like duplicates and missing values
- What has the EDA told you?
 - The dataset shows that 83.4% of employees stayed while **16.6%** left. Employees who resigned exhibited lower satisfaction levels. They also tended to have higher evaluation scores, participated in more projects, and logged more average monthly working hours than those who remained with the company.
 - Attrition at Salifort Motors is not driven by a single factor but by a combination of workload, tenure, and lack of growth. By looking at the distributions and relationships for and between these, we can identify specific "risk profiles" for attrition.
 - The data reveals that employees who leave generally fall into two distinct categories:



- Overworked High-Achiever*: These employees have high evaluation scores, work excessive hours (often exceeding 250–300 hours/month), and handle 6 or 7 projects. Every single employee in the dataset with 7 projects left the company. This is a massive red flag for workload management. Their attrition is likely due to burnout and is indicative of a significant High Performer Flight Risk.
- Disengaged and Under-utilized Under-performer: These employees work significantly fewer hours (around 150/month), have low satisfaction levels, and only handle 2 projects. Their attrition is likely due to lack of engagement or being managed out. Employees with only 2 projects have a surprisingly high exit rate (potentially feeling undervalued or underutilized).
- Employees who remain with the company exhibit a high degree of homogeneity across core metrics. This "retention cluster" is characterized by a sustainable equilibrium: performance evaluations between 0.60 and 0.85, a moderate workload of 160 to 230 monthly hours, and involvement in 3 to 4 projects. This balance directly correlates with healthy satisfaction levels, typically ranging from 0.55 to 0.85, suggesting that this specific workload-to-reward ratio is key to long-term employee stability.
- The organizational Danger Zone is the 3 to 5-year tenure window.
 - Attrition peaks sharply between 3 and 5 years
 - If an employee stays past the 6-year mark, the likelihood of them leaving drops significantly. This suggests that retention efforts should be hyper-focused on mid-tenure employees.
 - While projects, hours, and evaluations skyrocket, during the mid-tenure (3-5 years) critical period, the promotion rate is flat and near zero until Year 7. It is crucial to note that employees with a propensity to leave do not remain with the company beyond Year 6. This indicates that by the time the promotion window opens, the organization has already lost the cohort most likely to resign. This is a classic "too little, too late" management failure.
 - Targeted interventions—such as workload rebalancing for high-performers and career development for under-utilized staff—are most critical during this period to prevent the loss of both current productivity and future leadership.



- Mid-tenure employees (3–5 years) have significantly lower satisfaction levels compared to all other tenure groups. This result is statistically robust.
- Mid-tenure employees represent 93.0% of all resignees and the resignation rate within this group is 22.4%.
- The data suggests that turnover isn't just about "unhappy" people. There is a significant retention issue with high achievers. The High- working hour/High-Evaluation group amongst the resignees represents the most "expensive" loss to the company as it is losing its most competent talent.
- The current incentive structure at Salifort Motors is fundamentally misaligned, inadvertently penalizing top performers. By adhering to a "tenure-gated" promotion model (Year 7) rather than a "merit-based" framework (Year 4), the organization effectively rewards high performance (Evaluation > 0.75) with increased workload instead of career advancement. To stabilize the workforce, management must intervene by Year 3, decoupling high performance from infinite workload scaling. Transitioning to a model where high evaluations trigger promotions or salary adjustments—rather than additional projects—is essential to retaining high-value talent.
- What resources do you find yourself using as you complete this stage?
 - Coursera Course Materials and Labs: Videos, reading assignments, and glossaries. Python and Python Documentation. Gemini
- Do you have any ethical considerations in this stage?
 - Are there any potential biases about the type of people who leave? Anything related to gender, nationality, religion or race? Is the data biased? How do we avoid bias being injected and reproduced in our models?
 - How do we prioritize fairness in the data that we have and use? What steps are we taking to reduce the potential of unintended consequences of your machine learning applications? How do we minimize the risk of exposing the business and its employees to negative consequences due to bias? The negative consequences can be the result of bias in the training data, which was inherited from bias in the data collection process so how do we avoid this?



Data Project Questions & Considerations



PACE: Construct Stage

Foundations of Data Science

- Do any data variables averages look unusual?
 - The data for employee tenure contains some significantly high values (outliers).
 - Most people in the dataset were never promoted in the last 5 years
 - All employees who were assigned 7 projects had left
 - All employees who resigned did so before their 7th year
- How many vendors, organizations or groupings are included in this total data?
 - One Organization 10 Departments,, 3 salary levels,

Go Beyond the Numbers: Translate Data into Insights

- What data visualizations, machine learning algorithms, or other data outputs will need to be built in order to complete the project goals?
 - Histograms, Boxplots, Scatterplots, BarCharts, Heatmap, Confusion Matrix, Feature Importance
 - Logistic Regression, Random Forest Model, XGBoost Model
- What processes need to be performed in order to build the necessary data visualizations?
 - Data Cleaning and Processing (Feature Engineering), constructing relevant data frames and groupbys
- Which variables are most applicable for the visualizations in this data project?
 - Satisfaction Level
 - Average Monthly Hours
 - Tenure
 - Last Evaluation
 - Projects Contributed (Number of Projects)
- Going back to the Plan stage, how do you plan to deal with the missing data (if any)?
 - N/A as the data had no missing values



The Power of Statistics

- How did you formulate your null hypothesis and alternative hypothesis?
 - Null Hypothesis: No difference between Mid-Tenure and Outside of Mid-Tenure employees when it comes to Satisfaction Levels / Willingness to resign
 - Null Hypothesis: There is a difference Mid-Tenure and Outside of Mid-Tenure employees when it comes to Satisfaction Levels / Willingness to resign
- What conclusion can be drawn from the hypothesis test?
 - A Welch's t-test confirmed that mid-tenure employees (3-5 years) have significantly lower satisfaction levels compared to all other tenure groups ($t = -15.63$, $p < 0.001$). This result is statistically robust and represents a critical "danger zone" for employee retention.

Regression Analysis: Simplify Complex Data Relationships

- Do you notice anything odd?
 - The significant influence of the Satisfaction Level variable may be overshadowing the effects of other predictors.
 - Low recall and F1 score for Logistic Regression
- Can you improve it? Is there anything you would change about the model?
 - Drop Satisfaction Level

The Nuts and Bolts of Machine Learning

- Is there a problem? Can it be fixed? If so, how?
 - We can drop Satisfaction Level and use Boosting and Bagging to avoid overfitting
- Which independent variables did you choose for the model, and why?
 - I used all except for Tenure (created a mid_tenure variable instead) and Satisfaction Level
 - Maximum variables kept to avoid bias
- How well does your model fit the data? (What is my model's validation score?)
 - XGBoost Test (Champion):
 - Precision: 0.871314
 - Recall: 0.816583
 - **F1: 0.843061**
 - Accuracy: 0.949562



- Can you improve it? Is there anything you would change about the model?
 - XGboost Validation model had a high enough score for me to proceed to use it as the champion model
- Do you have any ethical considerations in this stage?
 - Are there any potential biases about the type of people who leave? Anything related to gender, nationality, religion or race? Is the data biased? How do we avoid bias being injected and reproduced in our models
 - How do we prioritize fairness in the data that we have and use? What steps are we taking to reduce the potential of unintended consequences of your machine learning applications? How do we minimize the risk of exposing the business and its employees to negative consequences due to bias? The negative consequences can be the result of bias in the training data, which was inherited from bias in the data collection process so how do we avoid this?
 - How is the model's error distributed over a population? If the model only makes errors in particular cases that are similar, it could carry higher ethical risk.
 - The lower precision score of the Logistic Regression model could lead to inefficient allocation of resources if used for decision-making, as the company might focus on retaining employees who are not actually at risk of leaving.



Data Project Questions & Considerations



PACE: Execute Stage

Foundations of Data Science

- Given your current knowledge of the data, what would you initially recommend to your manager to investigate further prior to performing an exploratory data analysis?
 - Are women or men more likely to leave?
 - Are younger or older employees more likely to leave?
 - Did the employee receive a bonus?
 - Are new parents more likely to leave?
 - What's the geographic location of the employees?
 - How do the benefits offered by the company compare to the industry?
 - Can we get data from outside sources e.g. average market salary for employees role and department and see how the current salary compares to it?
 - What was the month and year of resignation?
 - Data from exit interviews: did they have a counter offer?
- What data initially presents as containing anomalies?
 - Tenure
- What additional types of data could strengthen this dataset?
 - Demographic variables (Gender, Age, Is an Employee a New Parent, Location?)
 - Bonus Received as a proportion of Salary
 - Benefits offered in Industry vs those offered by company
 - Average Market Salary for Given Role
 - Data from exit interviews: did they have a counter offer?

Go Beyond the Numbers: Translate Data into Insights

- What key insights emerged from your EDA and visualizations(s)?
 - See answer above for the Analyze Stage



- What business recommendations do you propose based on the visualization(s) built?
- Given what you know about the data and the visualizations you were using, what other questions could you research for the team?
- How might you share these visualizations with different audiences?
 - [Executive Summary, Google Slides, PDF](#)

The Power of Statistics

- What key business insight(s) emerged from your A/B test?
 - [A Welch's t-test confirmed that mid-tenure employees \(3-5 years\) have significantly lower satisfaction levels compared to all other tenure groups \(\$t = -15.63, p < 0.001\$ \). This result is statistically robust and represents a critical "danger zone" for employee retention.](#)
- What business recommendations do you propose based on your results?
 - [See business recommendations below](#)

Regression Analysis: Simplify Complex Data Relationships

- To interpret model results, why is it important to interpret the beta coefficients?
 - [Beta coefficients tell us the magnitude and direction of change](#)
- What potential recommendations would you make to your manager/company?
 - [See business recommendations below](#)
- Do you think your model could be improved? Why or why not? How?
 - [Yes](#)
 - [By running ML models](#)
- What business recommendations do you propose based on the models built?
 - [Cap Project Assignments:](#)
 - [Limit the number of projects per employee to a maximum of 5. EDA shows a massive spike in turnover at 6 projects.](#)
 - [Monitor Work Hours:](#)
 - [Implement alerts for managers when an employee exceeds 200–210 hours per month consistently.](#)
 - [Focus on employees in the 3–5 year tenure bracket](#)



- Management should take proactive steps to ensure employees in the 3–5 year tenure bracket are adequately supported and recognized. It is essential that high-performing employees feel valued; therefore, exceptional performance evaluations should be directly linked to clear career advancement pathways or financial incentives. As an immediate next step, the company should conduct focus groups with employees in this high-risk cohort to gather qualitative insights and develop data-informed strategies for improving job satisfaction.
- Investigate Low-Engagement Groups:
 - Address why employees with only 2 projects have low satisfaction; they may need more challenging work or better integration into the team.
- What key insights emerged from your model(s)?
 - Satisfaction level, evaluation score and being in the mid-tenure employee group have the highest coefficients in the model, meeting our expectations based on the EDA and our high performer flight risk theory.
 - As expected, the negative coefficients for `satisfaction\_level` indicates that lower satisfaction is associated with higher log-odds of resignation.
 - Additionally, the positive coefficient for `latest\_evaluation\_score` indicates that higher evaluation scores are also associated with higher (not lower) log-odds of resignation. This is in line with our high performer flight risk theory as it suggests that employees with higher evaluation scores are more likely to resign either due to burnout from being overworked or because they find better opportunities elsewhere due to their strong performance.
 - This could also be because they feel more confident in their ability to find a new job.
 - The positive coefficient for `mid\_tenure` also indicates that being in the mid-tenure group is associated with higher log-odds of resignation compared to those outside of that group, which is in line with our EDA findings and our theory about the critical retention period of 3-5 years.
 - It is interesting to note that the `average\_monthly\_hours` predictor does not make it to the top 3. This could be because its impact may already be captured by the `satisfaction\_level`, `latest\_evaluation\_score` and `mid\_tenure` predictors.
- Do you have any ethical considerations at this stage?
 - High False Positive Rate: The model is quite "noisy." It flags almost one third of actually Retained employees as potential resignations (these people actually had no intention of leaving). In a business context, this could lead to wasting retention resources on the wrong people.

The Nuts and Bolts of Machine Learning

- What key insights emerged from your model(s)?
 - The dataset shows that 83.4% of employees stayed while **16.6%** left. **The most frequently reported satisfaction level for employees who resigned is the lowest value for satisfaction (~ 0.1).** Remarkably, These employees also tended to have higher evaluation scores, participated in more projects, and logged more average monthly working hours than those who remained with the company.



- The story of Salifort Motors is one of unbalanced excellence. The company is successfully identifying and utilizing its "Champions" (the high-project, high-evaluation group), but it is failing to protect them from burnout. To change the ending of this story, the focus shouldn't just be on the "bad" employees leaving, but on recalibrating the workload for the "best" ones before they reach their mid-tenure exit ramp.
- Through EDA, several critical patterns emerge regarding employee turnover. Results from the Logistic Regression and Machine Learning Models confirmed many of these insights the EDA.
- The ****Overworked Burnout/High Performer Flight Risk****:
 - EDA showed that there is a strong correlation between high average_monthly_hours, higher evaluation, high number_project (6 or more), and attrition. Employees working over 240 hours per month are significantly more likely to leave. Satisfaction levels drop drastically for employees assigned to 6+ projects or those working 260 or more hours.
 - The XGBoost model also predicted that the company's top performers—those taking on the most projects, working the most hours, and receiving the highest marks—are the ones most likely to walk out the door. These insights suggest a "reward" system where high performance is met with an unsustainable workload, leading to a "breaking point" for the most talented staff.
- The ****Mid-Tenure Danger Zone****:
 - EDA showed that employees with 3–5 years of tenure (time_spend_company) show the highest turnover rates. Those who stay beyond 6 years are much more likely to remain, suggesting a critical period for retention efforts.
 - There is a strong consensus between both the Logistic Regression and the XGBoost models regarding the significance of `mid\_tenure`. Aside from the dominant effect of satisfaction level in the Logistic Regression model, `mid\_tenure` maintains its position as the second most significant predictor in both modeling approaches. Employees aren't just leaving in their first few months (the honeymoon phase) or after many years (the invested phase). Instead, there is a "danger zone" in the middle of their tenure. At this stage, employees have learned the ropes and contributed significantly, but they may feel they've hit a ceiling or are being overlooked for the next step, especially if they haven't seen a promotion in the last five years—a factor Logistic Regression identifies as a contributor to attrition.
- The Disengaged Under-performer:
 - These employees work significantly fewer hours (around 150/month), have low satisfaction levels, and only handle 2 projects. Their attrition is likely due to lack of engagement or being managed out. Employees with only 2 projects have a surprisingly high exit rate (potentially feeling undervalued or underutilized).
 - **Promotion & Recognition****: A lack of promotions in the last 5 years combined with high performance evaluations (last_evaluation) is a common trait among those who left, suggesting that high-achieving employees feel undervalued. The XGBoost model



includes promotion in it's top 10 predictors with the Logistic Regression Model making it the top 5th predictor

- What are the criteria for model selection?
 - F1 score
- Does my model make sense? Are my final results acceptable?
 - Yes
- Were there any features that were not important at all? What if you take them out?
 - Yes Department
- Given what you know about the data and the models you were using, what other questions could you address for the team?
 - To facilitate a more granular and robust subsequent analysis, we recommend incorporating the following inquiries into future survey cycles. These additional data points will provide the necessary depth to refine our predictive models and further isolate the drivers of attrition.
 - Are women or men more likely to leave)
 - Are younger or older employees more likely to leave?
 - Did the employee receive a bonus?
 - Are new parents more likely to leave?
 - What's the geographic location of the employees?
 - How do the benefits offered by the company compare to the industry?
 - Can we get data from outside sources e.g. average market salary for employees role and department and see how the current salary compares to it?
 - What was the month and year of resignation?
 - Data from exit interviews: did they have a counter offer
- What resources do you find yourself using as you complete this stage?
 - Coursera Course Materials and Labs: Videos, reading assignments, and glossaries. Python and Python Documentation. Gemini
- Is my model ethical?
 - Yes
 - In concluding this analysis, it is imperative that the following ethical considerations serve as a foundation for any subsequent decision-making and policy implementation



- Demographic Disparity: Does the attrition data indicate disparate impact across protected groups, such as gender, ethnicity, religion, or race?
 - Algorithmic Bias Mitigation: What is the risk of systemic bias within the dataset, and what strategies will be employed to prevent the model from reinforcing these historical inequities?
 - Fairness Frameworks: How can we operationalize fairness within our data pipeline to ensure equitable treatment of all employee segments?
 - Risk Management: How can we establish audit procedures to minimize the negative consequences for both the organization and its employees resulting from biases inherited during data collection?
- When my model makes a mistake, what is happening? How does that translate to my use case?
 - Yes, a wrong prediction could lead to a wastage of resources or missed opportunities